ANNOTATIONES ZOOLOGICAE JAPONENSES
Volume 26, No. 2—June 1953

Published by the Zoological Society of Japan
Zoological Institute, Tokyo University

Studies on the Mechanism of Pigment Migration within Fish Melanophores with Special Reference to Their Electric Potentials[1) 2)]

With 4 Text-figures

Haruo KINOSITA

Zoological Institute, Faculty of Science, Tokyo University
(*Communicated by* T. YAMAMOTO)

A number of investigators (Ballowitz '14, Schmidt '21, Biedermann '26, Matthews '31, Kamada and Kinosita '44) have shown that the so-called contraction and expansion of the dermal melanophore of adult fish is due to an intracellular migration of pigment granules and is not attributable to an amoeboid activity of the cell. Very little is known, however, about the exact nature of the mechanism of pigment migration. According to Zimmermann ('93), Ballowitz ('14), Schmidt ('21) and Perkins and Snook ('32), mechanical pressure generated locally in the cell drives the cell contents into or out of the centrosphere. This possibility is strongly opposed by the fact that granules within melanophore processes isolated from the centrosphere are still capable of concentration and dispersion (Matthews '31), even if both end parts of the isolated process are cut off (Kamada and Kinosita '44). As is pointed out in the previous paper (Kamada and Kinosita '44), so-called contraction-inducing agents, such as KCl or adrenine, cause the granules in the isolated branch to migrate in the direction which originally led to the centrosphere, and form a clumped mass at the proximal extremity of the cut branch, contrary to the result of Matthews ('31), who describes...."the pigment peripheral to the injury clumped itself near the middle of the isolated process" (p. 480). This discrepancy is significant, because our observation cannot be explained

1) Read before the 23rd Annual Meeting of the Zoological Society of Japan, Oct. 1952.

2) The expense of this research has been defrayed from a Governmental Grant in Aid for Fundamental Scientific Research.

simply by the colloidal theory, as postulated by Spaeth ('16) and Matthews ('31), or by an hypothesis assuming intracellular contractile structures carrying the pigment granules (Ballowitz '14, Franz '39 and Marsland '44). Furthermore, the polarity observed in the direction of "proximal" and "distal" migration of granules within the isolated process may be taken to suggest the existence of some physiological gradient along each process of the melanophore.

The present investigation was undertaken to obtain information on the nature of such a gradient. In this connection it is important to refer again to our foregoing paper, in which it was shown that the melanin granules are negatively charged, because, under certain conditions, the intracellular granules as well as those which have burst out of the cell may be forced to migrate toward the anode by sending direct current. Therefore, it becomes desirable to study the local differences of electric potentials within a single melanophore, in order to determine whether or not the intracellular migration of the granules may be explained by the assumption of electrophoresis.

I

A scale of *Oryzias latipes* was carefully removed from the animal, and was fixed, epidermis side down, under a cover glass as is shown in Fig. 1 (above). For microscopic observation of the response of the scale melanophore (diameter in expanded state: $80 \sim 120\mu$) and, at the same time, for determination of its electric potential, the cover glass was mounted on a glass trough ($5 \times 5 \times 40$ mm), specially constructed on a mechanical stage (Fig. 1, below). The epidermal surface of the scale was kept immersed in physiological solution (M/7.5 NaCl, M/7.5 KCl and $2/3 \times$ M/7.5 $CaCl_2$, mixed in a volume ratio of 100 : 2.0 : 2.1 respectively, pH adjusted to 7.3 by $NaHCO_3$), which was made to flow through the trough at a constant rate (0.8 mm/sec), so as to minimize the thermal effect due to the illumination. Contraction and expansion of the melanophore were induced, unless otherwise stated, by changing the physiological solution to isotonic (M/7.5) KCl solution (pH=7.3 by $NaHCO_3$) and vice versa.

Microelectrodes for determination of the electric potentials were made of thin quartz capillary tubing filled with 3M KCl solution connected with AgCl-Ag, having tip diameters of about 2μ and electric resistances of $2 \sim 3$ megohms. They were held in the running fluid just beneath the scale, their positions being adjusted correctly by means of micromani-

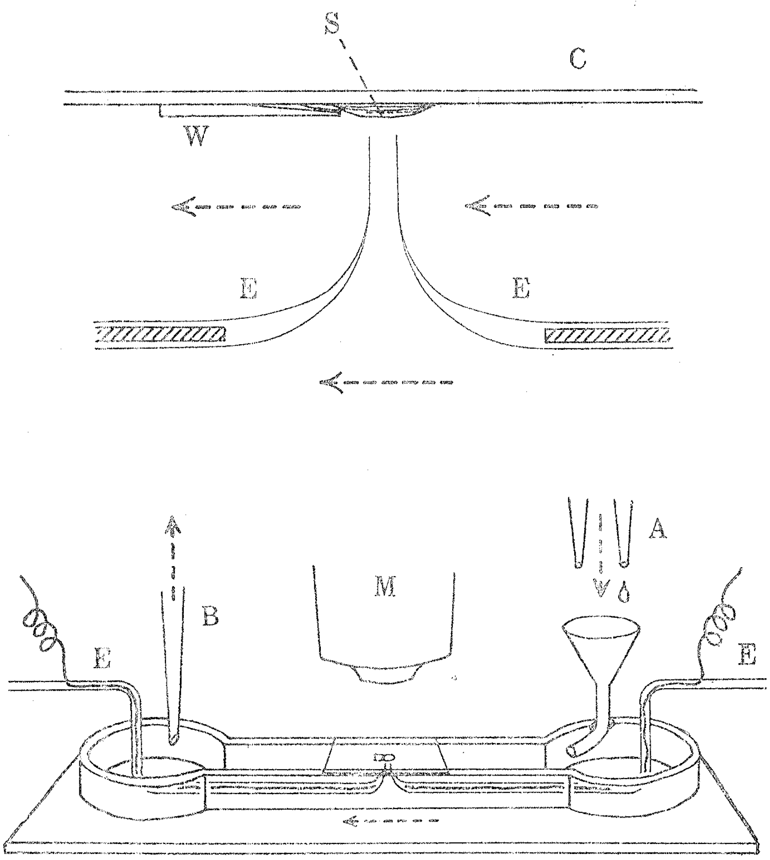

Fig. 1. Diagrammatic representation of the experimental arrangement.
above: Lateral view of the scale and the capillary microelectrodes.
below: Glass trough constructed for the determination of the electric potentials by the capillary microelectrodes inserted into the melanophore under the effect of various external solutions.
A: inflow siphons connected with the solution reservoirs, furnished with stoppers and screw cocks.
B: outflow pipette connected with suction pump.
C: cover glass.
E: quartz capillary microelectrodes connected with micromanipulators.
M: microscopic lens.
S: single piece of scale.
W: wedge-shaped piece of cover glass to hold the scale in position.
Arrows in broken lines indicate the flowing directions of experimental solutions.

pulators; one of them was inserted vertically into a melanophore[1)], while the other, kept at a distance, served as an independent electrode. Electric potentials between these electrodes were read on a sensitive pointer galvanometer connected with a two-step direct-coupled amplifier having an input resistance of more than 10^3 megohms. Fluctuations of potential

1) The melanophores are often found superimposed on iridocytes. In the present experiment, however, melanophores existing in the close vicinity of iridocytes or other structures like xanthophores and blood vessels were excluded from the electric measurements.

values due to flow or renewal of the solutions or to the displacement of electrodes were found to be negligible.

II

(1) If one of the microelectrodes is lifted slowly through the surface layers of the scale to penetrate the melanophore, a definite change occurs in the electric potentials as is shown in Fig. 2. The first sign of electric change appears when the tip of the electrode is introduced into the surface tissue, after coming in contact with the optically discernable scale surface. The introduced electrode becomes gradually electro-negative with respect to the indifferent one with further insertion. The slow change is succeeded by a sudden increase when the melanophore is impaled with the electrode, the moment of impalement being clearly recognized by a quick dispersion of melanin granules from the inserted spot. Further upward shifting

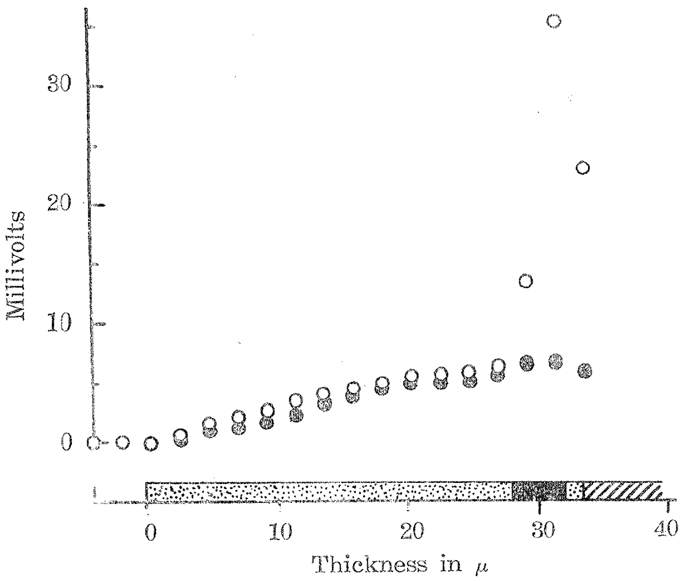

Fig. 2. Development of negative potentials with gradual insertion of microelectrode into the scale skin.

Hollow circles: case of insertion through the surface layers into the melanophore.
Solid circles: case of insertion into the skin of the same scale free from chromatophores or blood vessels.
Below: position of melanophore in cross section of the same scale.
Dotted part: layers of epidermal cells.
Solid part: thickness of the melanophore.
Striated part: horny part of the scale.

Electrode is introduced into the centrosphere of a melanophore expanded in physiological solution. Diameter of the electrode tip: 1.8 μ. Diameter of melanophore: about 90 μ. Diameter of centrosphere: about 20 μ. Temp.: 21.0°C.

of the electrode causes an abrupt decrease of negativity, probably because the electrode tip penetrates the upper membrane of the melanophore. In fact, such a melanophore shows, after removal of the electrode, a small clear perforation at the inserted point irrespective of the intracellular movement of pigment granules. In many cases it is observed that the lifted electrode is bent by striking against a solid surface at the moment of potential decrease. Therefore, it seems likely that the perforation is induced by the electrode coming against the hard horny part of the scale existing closely above the melanophore.

This explanation of the sequence of events may be supported by the following observations. When the above-mentioned measurement is over, the scale just used is detached from the cover slip, and is severed by fine scissors under a binocular microscope in the approximate vicinity of the observed melanophore. Thus, by appropriate manipulation, it is possible to observe the cross section of the scale in the living condition under the microscope. The thickness of the surface layers as well as the position and thickness of the melanophore are determined, and the data are superimposed on the foregoing result, as is shown in the lower part of Fig. 2. The striking coincidence of the first and the second rising points of the curve with the level of the scale surface and that of the melanophore respectively clearly indicates that the first slow potential change is due to the surface layers, while the second rapid increase is attributable to the melanophore.

As a matter of fact, when a microelectrode is inserted into the chromatophore-free portion of the scale skin in the immediate neighbourhood of the measured melanophore, only a slow curve, quite similar to that recorded previously, is obtained, as is represented by the solid circles in the same figure. Furthermore, shifting of the microelectrode toward the horny part of the scale completely lacking in integument cells or toward skin severely injured by stabbing with a microneedle never gave rise to a detectable change of potential.

It was found that the larger potential due to the melanophore, which will be hereafter called the "melanophore potential" for convenience, is largely dependent on the melanophore activity as well as on the position of the inserted electrode, whereas the minor potential of the skin tissue is proved to be practically unaffected by the change of medium causing the melanophore response. Melanophore potentials measured by a microelectrode inserted into the centrosphere and into thick projections of both expanded and contracted melanophores are given in Table 1.

Insertions are made into the centre of centrospheres (diameter: $18\sim24\,\mu$), or into the relatively thick projections (width: $13\sim20\,\mu$) at points $19\sim32\,\mu$ distant from the centre of the centrosphere. A minor variation of electrode diameter does not seem, at least qualitatively, to modify these results, although insertions of an electrode as thick as several μ fail to detect any melanophore potential, probably because of the injurious effect of impalement. Observation of emptied projections in a contracted melanophore is very difficult, as is pointed out by Matthews ('31). Therefore, measurement of the melanophore potential at the invisible branch of a contracted melanophore is made by lifting a microelectrode up to a point, indicated by an ocular micrometer, which has previously been determined, in the expanded state, to be a point on one of the thick projections of the melanophore.

Table 1.

Values of melanophore potentials.

Solution applied	State of melanophore	Position of inserted electrode	Melanophore potential in m. v.	Number of determinations	Temperature °C
Physiological solution	expanded	centrosphere	-33.1 ± 0.2	47	27.5~30.0
		projection	-29.7 ± 0.3	23	28.5~29.0
0.00025 M atropine sulfate in physiological solution	expanded	centrosphere	-32.3 ± 0.5	14	27.5~28.0
		projection	-29.5 ± 0.6	10	27.5~28.5
Isotonic KCl solution	contracted	centrosphere	-19.9 ± 0.2	51	27.5~30.0
		projection	-33.6 ± 0.4	20	28.5~29.0
0.00006 M adrenine in physiological solution	contracted	centrosphere	-25.5 ± 0.5	23	27.0~27.5
		projection	-37.1 ± 0.7	18	27.0~28.0

As is shown in the table, the values of the negative potentials of expanded melanophores are always larger at the centrospheres than at the projections, while those of contracted ones are apparently smaller at the centrospheres than at the projections, regardless of the nature of the agents used to induce expansion or contraction. This fact suggests that, in the expanded state, the interior of the centrosphere is electrically more negative (in the internal circuit) than that of the peripheral branches, and this gradient of electric potential is reversed by the action of contraction-inducing agents, such as KCl and adrenine.

(2) The electric sign of the melanin pigments is examined by observing the direction of pigment migration when D.C. is sent through a pair of the same capillary microelectrodes inserted into the melanophore. As is mentioned in the previous paper (Kamada and Kinosita '44), it is shown that the granules in an expanded melanophore treated with 0.00025 M atropine sulfate clearly migrate toward the anode with a velocity depending on the voltage applied. When stimulated with an inserted microelectrode at suitable intensities, melanophores, expanded in physiological solution, also show little effect of indirect stimulation (cf. Kamada

and Kinosita '44), and the granules around the anode concentrate toward the electrode, while those around the cathode disperse. Also the granules within melanophores contracted in KCl or adrenine solution may be forced to migrate anaphoretically toward or from the microelectrode introduced into the clumped mass of pigments. The proximad or distad migration of the pigment granules, observed within the contracting or expanding melanophores, may be accelerated or retarded by electric current passed likewise through the inserted microelectrodes.

In all the cases mentioned above, it has been shown that the proximally migrating granules or those already clumped into a dense central mass show relatively little tendency toward electrophoresis, in comparison with the granules in the fully expanded melanphores or those undergoing distad migration, showing marked Brownian movement. Moreover, most of the granules, which migrate only slightly toward the anode during the closure of the circuit, move shortly backward on breaking the current, as is pointed out briefly in the foregoing paper (Kamada and Kinosita '44). These observations give the impression that the protoplasm of expanding or expanded melanophores is rather in a state of sol, while that of the contracting or contracted ones has more or less a gel structure, as is suggested by Matthews ('31) and Marsland ('44).

At any rate, it is shown that the melanin granules always migrate toward the anode, or are negatively charged. This may be ascertained by the fact that if erythrocytes, which are generally known to be negatively charged (e.g. Abramson '34), are occasionally located near the melanophore, they are forced to migrate, within the capillary vessel, in the same direction as that of the melanin granules at the moment of closure of the circuit. The electrophoretic behavior of the pigment granules within the melanophore was already referred to by Beauvallet and Veil ('32), who observed a small protrusion of melanin granules toward the anode and their withdrawal at the cathodal side when a contracted melanphore was stimulated with D.C. However, instead of interpreting this as a result of electrophoresis, the authors seemed to emphasize the significance of polar excitation.

(3) Since it has been proved in the preceding sections that there is a gradient of electric potential along the radial axis of the melanophore, which may be reversed by the action of contraction-inducing agents, and that granules within the cell are electrically charged, it becomes desirable to follow the time-change of the melanophore potential in close connection with the movement of the pigment granules.

Melanophore potentials were measured between a pair of capillary microelectrodes, one of which was inserted into the centrosphere of a melanophore, the position of the inserted electrode being adjusted to give rise to a just maximal melanphore potential. At the same time, the length of one of the branches of the melanophore, into which the microelectrode is introduced, was determined by means of an ocular micrometer. The values of the electric potentials as well as of the lengths of a branch were determined on a single melanophore throughout a series of alternations of the bathing fluid from physiological solution to isotonic KCl solution and vice versa. In the most successful cases, it was possible to reproduce the change several times with little shift of the initial levels of potential and length. One of the typical examples is shown in Fig. 3.

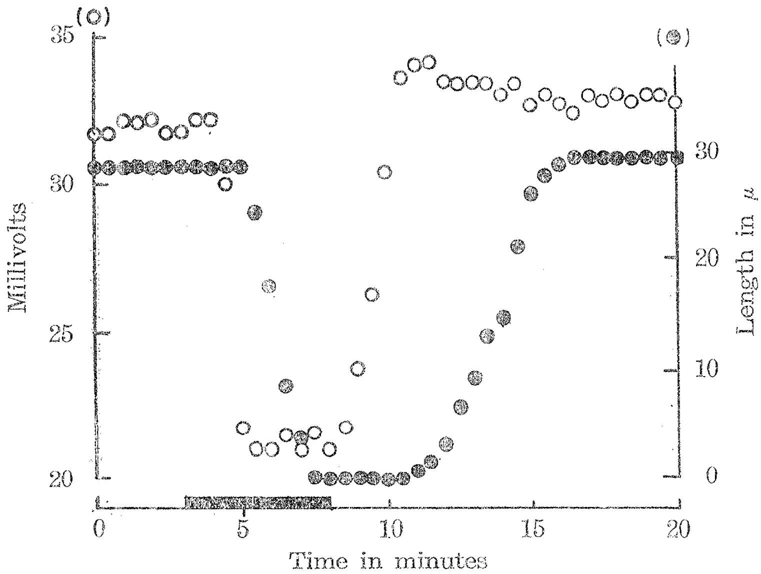

Fig. 3. Result of simultaneous determinations of change in melanophore potential (hollow circles) and the length of process (solid circles) of the same melanophore when the medium is altered from physiological solution to isotonic KCl solution (solid rectangle) and again to physiological solution. Microelectrode is introduced into the centrosphere of the melanophore. Diameter of the electrode tip: 2.0 μ. Temp.: 21.5°C.

Change of the medium from physiological solution to KCl solution causes an abrupt decrease of the negative potential at the centrosphere followed by a retraction of the melanophore projection. Obviously the melanin granules, which have been proved to be negatively charged, are carried by the electric current toward the centrosphere, where the electric potential is relatively raised by the action of the KCl. The decreased state of melanophore potential at the centrosphere, which may be expected from the result shown in Table 1, is continued as long as the melanophore is kept contracted in the KCl solution. On return to physiological solution, however, the melanophore potential and later the length of melanophore projection recover their initial levels, suggesting, as before, the possibility of electrophoresis toward the new anode.

The time delay of the onset and recovery of these responses after the moment of the change of medium, which are indicated in the figure by the two ends of the black rectangle on the abscissa, may be taken to represent the time required for the attainment of an effective concentration of ions at the melanophore surface plus the reaction time of the melanophore. It is important to note here that the first signs of decrease and of recovery of the melanophore potential, especially the latter, always precede the beginning of the pigment granule migration. Although, at the present state of the investigation, it is impossible to give a satisfactory explanation for the delay, it seems probable that the change of colloidal state of the protoplasm may act as one of the limiting factors of the pigment displacement. However, it is at least evident from the precedency of the electric change that the electric potential is not the result but more probably the cause of the migration of charged pigment granules.

In the next attempt both the mircroelectrodes were introduced into a melanophore, one into the centrosphere, and the other into a thick projection of the same melanophore. The potential difference as well as the length of the projection were determined as before. One of the representative results is given in Fig. 4.

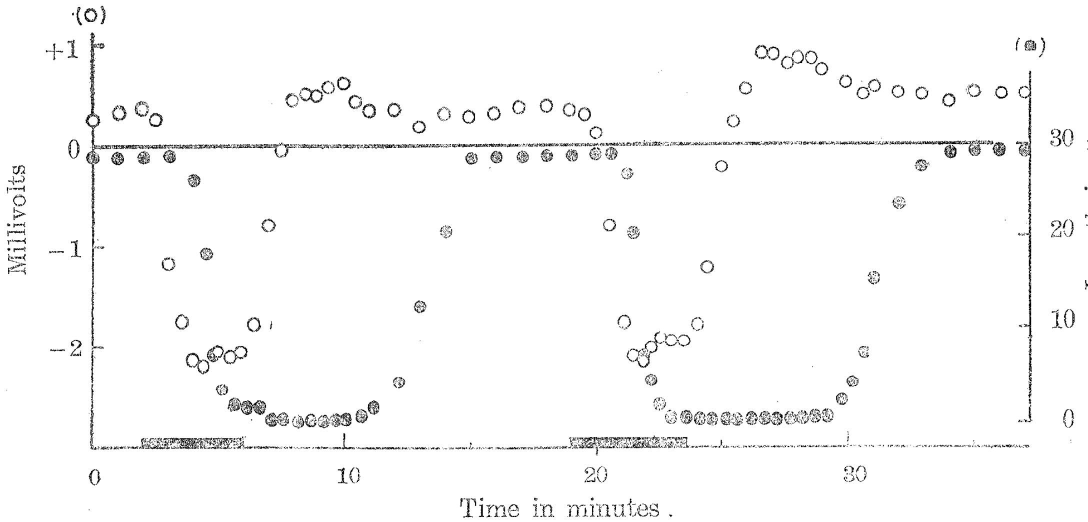

Fig. 4. Same as Fig. 3, except that both microelectrodes are introduced into the melanophore; one (tip diameter: 1.8 μ) into the centrosphere, the other (tip diameter: 1.5 μ) into the branch (width: 13 μ). Distance between electrodes: 35 μ. Ordinate: values of potential negativity of centrosphere relative to the peripheral branch. Temp.: 22.5°C.

The polarity of the electric potential, which is initially more negative (in the internal circuit) at the centrosphere, may be reversed and then restored repeatedly by alternative applications of isotonic KCl and physiological solution. Although the potential values are apparently smaller than those to be expected from Table 1, probably as a result of the damage caused by the impalement with two electrodes, it is clearly shown in the figure that the direction of pigment migration is closely associated with the polarity of the gradient of the electric potential, suggesting the electrophoretic behavior of the pigment granules.

III

If one may assume the electrophoretic migration of pigment granules within the melanophore, it is possible, at least qualitatively, to account for some of the results which have not been adequately explained heretofore.

(1) As is described in the previous paper (Kamada and Kinosita '44, p. 487), it is usually observed that, if a branch or centrosphere of an expanded melanophore is injured with a microneedle, the granules close to the injury tend to be concentrated at the cut end. The injured surface is expected to be electro-negative (in the external circuit) with respect to the intact surface of the cell, judged from the sign of the melanophore potential, so that it is natural that the negatively charged pigment granules in the neighbourhood of the injury are driven within the cell toward the cut end by a sort of injury current flowing in a similar direction as in the case of muscles and nerves.

(2) Within the process isolated from the centrosphere, the original polarity in the direction of concentration and dispersion of the pigment granules is still retained, as is mentioned already, even if the distal ends of the process are cut off. This fact, which does not seem to find a simple explanation in the hypothesis of sol-gel transformation or of contractility, may be smoothly interpreted by the above assumption that the charged pigment granules are driven by the electric current due to the gradient of electric potential along the melanophore processes.

(3) The velocity of centripetal migration of granules within the contracting melanophore is augmented if a cut is made by a microneedle in the proximal part of the branch, but it is reduced by a cut made at the distal end. On the other hand, the centrifugal migration is accelerated by making a cut at the distal end of the branch, and is retarded when the branch is injured at the proximal part (refer to Tables in pp. 488～490 of Kamada and Kinosita '44). These effects of injury may well be explained as acceleration and retardation of electrophoresis caused by the injury current which is expected to flow against, or to be superimposed on the electric current due to local difference of melanophore potentials, although quantitative analysis of these relationships depends on further investigations.

(4) According to micro-cinematographic studies, it often happens, during the repetition of rhythmic contractions and expansions, that granules within one branch are migrating distally, while, at the same moment, those in another branch of the same melanophore are migrating proximally (Kamada and Kinosita '44, p. 492). Moreover, as is pointed out

by Matthews ('31) and Kamada-Kinosita ('44), even the pigment granules within a single process of a melanophore may migrate quite independently. It seems possible to account for such behavior of the granules by local changes of the melanophore potentials of the branches or within a single branch. As a matter of fact, readings of melanophore potentials were not always stable and showed frequent fluctuations, especially in those aged melanophores showing a tendency to spontaneous activity.

(5) In the scale of *Oryzias*, melanophores and iridocytes behave quite oppositely, as is reported by Miyoshi ('52). Iridocytes expand in isotonic KCl solution, and contract in physiological solution, while melanophores contract in the former, and expand in the latter. Application of D.C. with the capillary microelectrodes inserted into a iridocyte revealed that the guanin granules are negatively charged, as in the case of melanin granules. On the other hand, measurement of electric potentials of iridocytes by the procedure described above indicated, however, that the interior of the centrosphere of the iridocytes contracted in physiological solution is electrically less negative (in the internal circuit) than that of the peripheral branches[1], while this gradient of electric potential is reversed, as in the case of melanophores, when the iridocyte expands under the influence of KCl. Thus, it is highly probable that the migration of guanin granules within the iridocyte is again attributable to the phenomenon of electrophoresis. The opposite behavior of these two sorts of chromatophores seems, therefore, not to be due to a difference of electric sign of the charged pigment granules, but more probably to the situation that the gradients of electric potentials of the two kinds of chromatophores are always opposite to each other under the influence of the same solution.

The movement of protoplasmic granules during the galvanotropic response of amoeba has been attributed to electrophoresis of positively charged particles by Heilbrunn and Daugherty ('39). However, no information seems to have been given concerning the local difference of bioelectric potentials in the amoeba during normal locomotion in support of their opinion.

The antero-posterior difference in the electric potentials of slime mold has been recorded by Watanabe, Kodati and Kinoshita ('37). It was found that the bioelectric potential undergoes spontaneous changes in close association with the rhythmic flow of the plasmodium. However, the

1) For example, insertion of the microelectrode into the centrosphere of a contracted iridocyte gave the potential value of -17.5 m.v., while the analogous values measured at two branches of the same iridocyte were -30.0 and -31.2 m.v. (Temp.: 29.0°C).

former is supposed not to be directly responsible for the latter (Kamiya and Abe '50).

Although there are experiments which may be interpreted as favouring the present view of pigment migration within melanophores, it is certain that this is not an exclusive explanation. Changes of colloidal state are probably related in some way to the mechanism of granule displacement (Matthews '31), as is obvious from the delicate change of Brownian movement occurring in association with the melanophore activities. Another possibility, of the pigment granules being attached to and carried by the intracellular contractile filaments or frame-work, as is proposed by Franz ('39) and Marsland ('44), is also attractive in connection with the apparent difficulty of the free mobility of melanin granules frequently observed during the present experiment.

The linkage between these possibilities is undoubtedly very complex and will be the subject of future investigations.

SUMMARY

1. Capillary microelectrodes having tip diameters of about 2μ were inserted into the scale skin of *Oryzias latipes*, and the electric potentials of the melanophores were measured.

2. Impalement of the melanophore is always accompanied by a sudden increase of electric negativity at the inserted electrode. This electric potential of the melanophore, which is called "melanophore potential" for convenience, is dependent on the melanophore activities as well as on the position of impalement, suggesting that there is a radial gradient of melanophore potentials, which may be reversed under a condition causing displacement of pigment granules.

3. Intracellular electrophoresis induced by an electric current sent by microelectrodes inserted into the melanophore revealed that the melanin granules within the melanophore are negatively charged.

4. Simultaneous determination of melanophore potentials and of pigment displacement seemed to suggest the following possibilities: when the contraction-inducing agent is applied, the centrosphere becomes electrically more positive (in the internal circuit) than the peripheral branches, so that the negatively charged melanin granules are driven by the electric current toward the centrosphere. The radial polarity of the melanophore potential is reversed on return to physiological solution, and consequently a distal electrophoretic migration of the pigment granules takes place.

5. Several experimental results, which seem to support this explanation of electrophoretic migration, are discussed.

LITERATURE

Abramson, H. A., 1934. Electric Phenomena in Their Application to Biology and Medicine. New York.

Ballowitz, E., 1914. Arch. ges. Physiol., Bd. 157, S. 165.

Beauvallet, M. and C. Veil, 1932. Comp. rend. Soc. Biol., T. 110, p. 698.

Biedermann, W., 1926. Erg. Biol., Bd. 1, S. 1.

Franz, V., 1939. Ztschr. Zellforsch. mikr. Anat., Bd. 30, S. 194.

Heilbrunn, L. V. and K. Daugherty, 1939. Physiol. Zool., Vol. 12, p. 1.

Kamada, T. and H. Kinosita, 1944. Proc. Imp. Acad. Tokyo, Vol. 20, p. 484.

Kamiya, N. and S. Abe, 1950. Journ. Colloid Science, Vol. 5, p. 149.

Marsland, D. A., 1944. Biol. Bull., Vol. 87, p. 252.

Matthews, S. A., 1931. Journ. Exp. Zool., Vol. 58, p. 471.

Miyoshi, S., 1952. Annot. Zool. Japon., Vol. 25, p. 21.

Perkins, E. B. and T. Snook, 1932. Journ. Exp. Zool., Vol. 61, p. 115.

Schmidt, W. J., 1921. Arch. Zellforsch., Bd. 15, S. 269.

Spaeth, R. A., 1916. Journ. Exp. Zool., Vol. 20, p. 193.

Watanabe, A., Kodati, M. and S. Kinoshita, 1937. Botan. Mag. (Tokyo), Vol. 51, p. 337.

Zimmermann, K. W., 1893. Verhdl. anat. Geselsch., S. 76.